J1A
nuMezzanine_4x34

TCK 23
T00 32 T00
T01 31 T01
T02 30 T02
T03 29 T03
T04 28 T04
T05 27 T05
T06 26 T06
T07 25 T07
T10 21 T10
T11 20 T11
T12 19 T12
T13 18 T13
T14 17 T14
T15 16 T15
T16 15 T16
T17 14 T17
T20 9 T20
T21 8 T21
T22 7 T22
T23 6 T23
T24 5 T24
T25 4 T25
T26 3 T26
T27 2 T27
TX0 13
TX1 12 TX1
TX2 11 TX2
TX3 10

J1C
nuMezzanine_4x34

BCLK 91
B00 100 B00
B01 99 B01
B02 98 B02
B03 97 B03
B04 96 B04
B05 95 B05
B06 94 B06
B07 93 B07
B10 89 B10
B11 88 B11
B12 87 B12
B13 86 B13
B14 85 B14
B15 84 B15
B16 83 B16
B17 82 B17
B20 77 B20
B21 76 B21
B22 75 B22
B23 74 B23
B24 73 B24
B25 72 B25
B26 71 B26
B27 70 B27
BX0 81
BX1 80 BX1
BX2 79 BX2
BX3 78

J1E
nuMezzanine_4x34

VCC_TL 1 VCC_TL
VCC_BL 103
VCC_BR 69
VCC_TR 35
3V3_SELECT 34
5V_SELECT 68 GND
GND 33
GND

J1B
nuMezzanine_4x34

LCLK 125 LCLK
L00 134 LO_UNUSED
L01 133 LO_UNUSED
L02 132 LO_UNUSED
L03 131 LO_UNUSED
L04 130 L04
L05 129 L05
L06 128 L06
L07 127 L07
L10 123 L10
L11 122 L11
L12 121 L12
L13 120 L13
L14 119 L14
L15 118 L15
L16 117 L16
L17 116 L17
L20 111 L20
L21 110 L21
L22 109 L22
L23 108 L23
L24 107 L24
L25 106 L25
L26 105 L26
L27 104 L27
LX0 115
LX1 114 LX1
LX2 113 LX2
LX3 112

J1D
nuMezzanine_4x34

RCLK 57
R00 66 R00
R01 65 R01
R02 64 R02
R03 63 R03
R04 62 R04
R05 61 R05
R06 60 R06
R07 59 R07
R10 55 R10
R11 54 R11
R12 53 R12
R13 52 R1_UNUSED
R14 51 R1_UNUSED
R15 50 R15
R16 49 R16
R17 48 R17
R20 43 R20
R21 42 R21
R22 41 R22
R23 40 R23
R24 39 R24
R25 38 R25
R26 37 R26
R27 36 R27
RX0 47
RX1 46 RX1
RX2 45 RX2
RX3 44

ADDRESS A0–A31

U1A
MC68030RC25B

A2 A0 A16 B9 T11
C4 A1 A17 A10 T10
D13 A2 A18 C8 T15
D12 A3 A19 A9 T12
C13 A4 A20 B8 T13
C12 A5 A21 A8 T14
D11 A6 A22 B7 T17
B13 A7 A23 A7 T16
B12 A8 A24 A6 T20
C11 A9 A25 B6 T21
A13 A10 A26 A5 T22
C10 A11 A27 B5 T23
B11 A12 A28 A4 T24
A12 A13 A29 B4 T25
B10 A14 A30 A3 T26
A11 A15 A31 B3 T27

DATA D0–D31

U1B
MC68030RC25B

K13 D0 D16 N9 B17
K12 D1 D17 M8 B16
L13 D2 D18 N8 B15
M13 D3 D19 N7 B13
L12 D4 D20 M7 B14
K11 D5 D21 N6 B11
M12 D6 D22 M6 B12
L11 D7 D23 N5 B10
N13 D8 D24 M5 B07
M11 D9 D25 N4 B06
L10 D10 D26 N3 B03
N12 D11 D27 M4 B05
M10 D12 D28 N2 B01
N11 D13 D29 M3 B02
M9 D14 D30 L4 B04
N10 D15 D31 N1 B00

THREE-STATE BUS CONTROL

U1C
MC68030RC25B

J2 AS
K1 CBREQ
C2 CIOUT
M1 DBEN
K2 DS
D2 FC0
C1 FC1
D1 FC2
B1 RMC
L3 R/W
L2 SIZ0
K3 SIZ1

ALWAYS-DRIVEN OUTPUTS

U1D
MC68030RC25B

B2 BG
M2 ECS
E13 IPEND
D3 OCS
J13 REFILL
J12 STATUS

CYCLE-CRITICAL INPUTS

U1E
MC68030RC25B

F1 DSACK0
G2 DSACK1
G1 STERM
L1 CIIN
J1 CBACK
E2 AVEC
H1 BERR
H2 HALT

OPEN-DRAIN HOST CONTROL / CLOCK

U1F
MC68030RC25B

F12 RESET
H13 IPL0
G13 IPL1
G12 IPL2
H12 CDIS
F13 MMUDIS
A1 BR
C3 BGACK
E1 CLK

POWER: ALL 10 VCC / 14 GND PINS

U1G
MC68030RC25B

L6 VCC
K4 VCC
K10 VCC
H3 VCC
H11 VCC
F2 VCC
F11 VCC
D4 VCC
D10 VCC
C6 VCC
L5 GND
L7 GND
L8 GND
L9 GND
J3 GND
J11 GND
G3 GND
G11 GND
F3 GND
E3 GND
E11 GND
C5 GND
C7 GND
C9 GND

RESERVED: DO NOT CONNECT

U1H
MC68030RC25B

K5 NC
F4 NC
E10 NC
E12 NC
D5 NC

PWR_FLAG
GND

C1 100nF — At U1 L6
C2 100nF — At U1 K4
C3 100nF — At U1 K10
C4 100nF — At U1 H3
C5 100nF — At U1 H11
C6 100nF — At U1 F2
C7 100nF — At U1 F11
C8 100nF — At U1 D4
C9 100nF — At U1 D10
C10 100nF — At U1 C6
C11 330pF — At U1 C6
C12 330pF — At U1 L6
C13 330pF — At U1 K4
C14 330pF — At U1 D4
C15 330pF — At U1 H3

LO_UNUSED
R1 47k
GND
LO: L00, L01, L02, L03

R1_UNUSED
R2 47k
GND
R1: R13, R14

C16 10uF
Bulk: observe polarity

HOST CONTRACT
5V_SELECT grounded: 3V3_SELECT floating. CPU clock: 12.5–25 MHz: timing not yet qualified.
X1/X2: open-drain only; keep enabled, release HIGH, never drive HIGH. Internal host pulls used.
Preload all tied unused bank bits LOW before every host-output enable.
Respect byte lanes and bus-release delays; never drive always-output bank.
CPU TTL output-high and host keeper margins are NOT guaranteed. See design-notes.md.

CPGA128 / 5 V / full bus, burst and takeover / experimental direct interface

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File: 68030.kicad_sch

Title: Motorola MC68030RC25B DUT daughter board

Size: A2 | Date: 2026-09-13 | **Rev: 1.0**
KiCad E.D.A. 10.0.6-rc2 | Id: 1/1